Release Summary

Product Name: SlugRoute
Team Name: Team SlugRoute
Date: June 3, 2026

Key User Stories and Acceptance Criteria

The following key user stories define the core functionality of the current release and serve as a guide for system acceptance testing:

User Story 1: Interactive Course Mapping

- **As a** UCSC student,
- **I want to** search for a class and have its lecture and associated section locations plotted on an interactive map,
- **So that** I do not have to look up coordinates across multiple campus sites manually.
- **Acceptance Criteria:**
    1. Typing two or more characters in the search bar triggers autocomplete suggestions.
    2. Selecting a course displays a dynamic preview dropdown with details and section checklists.
    3. Clicking "Add to Map" renders the lecture and selected sections on the map using customized SVG markers (star for lectures, square for labs, and triangle for discussions).
    4. The map automatically adjusts its viewport boundaries to fit all newly plotted class locations.

User Story 2: Map Visibility and Filtering Controls

- **As a** student seeking visual clarity,
- **I want to** toggle the visibility of meeting types globally and individual course cards selectively,
- **So that** I can declutter the map view and isolate specific class sessions.
- **Acceptance Criteria:**
    1. The sidebar menu displays dedicated checkbox toggles for Lectures, Labs, and Discussions; unchecking a category immediately removes those markers from the map.
    2. Each active course card in the sidebar contains a visibility toggle (eye icon) button.
    3. Clicking a card's visibility toggle hides all map markers associated with that specific course, regardless of category.
    4. Toggling a course card to "hidden" dims the sidebar card's opacity and applies a grayscale filter to visually denote its inactive status.

5. If a category or individual course containing the active route destination is toggled off, the active route is automatically cleared.

User Story 3: Walking Route Navigation & Time Estimation

- **As a** student on foot,
- **I want to** generate sequential walking routes and walking time estimates between my location and my classrooms,
- **So that** I can plan my transit times across campus.
- **Acceptance Criteria:**
    1. The user can establish a starting point using browser-based GPS geolocation or by manually pinning a custom point on the map.
    2. Clicking a class marker displays an InfoWindow with a "Get Directions" action button.
    3. Selecting "Get Directions" requests a walking path from the starting point to the destination via a secure backend proxy to the Google Routes API.
    4. The walking path is plotted as a polyline on the map, accompanied by a floating stat bubble displaying estimated walking duration (in minutes) and total distance (in miles).

User Story 4: iCalendar Schedule Export

- **As a** student,
- **I want to** export my mapped schedule to an .ics file,
- **So that** I can sync my UCSC class locations and recurring times with external calendar applications.
- **Acceptance Criteria:**
    1. The "Sync to Calendar" button is enabled only when there is at least one active or saved course.
    2. Clicking the button makes a POST request to /api/schedule/export with the combined list of active and saved courses.
    3. The server generates an RFC 5545-compliant iCalendar stream containing recurring weekly events based on correct UCSC term dates.
    4. The browser initiates a direct download of the generated .ics file.

Known Problems and Limitations

The following items represent known bugs, omissions, or design shortcuts in the current release version:

1. **Hardcoded Term Date Suffixes:** In backend/calendar.go (line 91), the dates for academic terms (Winter, Spring, Summer, Fall) are mapped to hardcoded calendar windows (e.g., Spring 2026 is statically assigned from March 30 to June 12). While

functional for the current year, this design shortcut does not dynamically query the UCSC registrar for actual academic calendar shifts.

2. **Cross-Browser Mobile Download Variance:** Mobile web browsers on iOS (such as Safari) and Android (such as Chrome) handle stream downloads of raw .ics file types differently than desktop browsers. On some mobile devices, the generated file may display as raw text within a new tab rather than launching a standard file-saving prompt.
3. **Absence of Real-Time Scraper Health Monitoring:** The PISA scraping mechanism (written in BeautifulSoup) successfully parses the registrar page. However, it lacks an automated alerting mechanism if UCSC changes its page layout, meaning layout breakages would only be noticed via manual inspection of the log output.
4. **Omission of SQLite Indexes for Search:** The database queries in backend/main.go perform wild-card LIKE lookups on every keystroke without an explicit index on the course_code column. Under heavy concurrent loads, this could introduce minor retrieval latency.

Product Backlog (Follow-on Project)

The following high-priority user stories and optimizations are proposed for a subsequent engineering cycle:

1. **Automated Dynamic Term Boundaries:** Integrate an automated query to dynamically fetch UCSC's official academic calendar bounds, removing the static switch-case assignments in backend/calendar.go.
2. **Database Performance Indexing:** Add database-level schema migrations to index the course_code column in the SQLite courses table to ensure sub-millisecond autocomplete queries.
3. **Webcal Stream Subscriptions:** Implement a subscription-based webcal:// link endpoint on the Go backend, allowing users to subscribe to dynamic schedule updates directly within their calendar apps rather than importing a static .ics file.
4. **Scraper Integrity Health Check:** Develop an automated daily test script that parses a mock PISA page, verifying the parser's regex and HTML selectors are fully functional, and reporting alerts via email or webhook if errors occur.
5. **Physical Campus Path Validation:** Conduct on-foot field tests across UCSC paths (e.g., between Science Hill and Cowell College) to verify Geolocation accuracy and identify potential routing deviations under weak cellular signals.